MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5ADP1, 5ADP2, 5ADP7, AND 5ADP14

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus with post accelerator

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Eb3 Eb2	Ehk	Alt
Unit:		V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Ratio	Vdc	ft
Maximum:	All	6.9	0,-200	550	1,100	2,850	6,600	1.5	1.0	2.3	±180	50,000
Minimum:	P1	5.7	---	---	---	1,500	1,500	---	---	---	---	---
	P2, P7, P14	5.7	---	---	---	1,500	3,000	---	---	---	---	---
TEST CONDITIONS:	P1	6.3	Adjust	---	Focus	1,500	3,000	---	---	---	---	---
	P2, P7, P14	6.3	Adjust	---	Focus	2,000	4,000	---	---	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Qualification inspection								
1141	Pressure (implosion)	All		---	---	---	---	---	---
1002	Barometric pressure, reduced	All	87 ±4 mmHg	---	---	---	---	---	---
5111	Vibration	All		---	---	Width	---	1	mm
1331	Direct-interelectrode capacitance	All	g1 to all	---	---	Cg1	---	7.9	pF
			k to all	---	---	Ck	---	5.8	pF
			D1 to D2	---	---	C1D2	---	3.1	pF
			D3 to D4	---	---	C3D4	---	1.3	pF
			D1 to all except D2	---	---	CD1	---	6.1	pF
			D2 to all except D1	---	---	CD2	---	6.1	pF
			D3 to all except D4	---	---	CD3	---	5.0	pF
			D4 to all except D3	---	---	CD4	---	5.0	pF
5101	Neck and bulb alinement (electrostatic types)	All		---	---	dia	---	2.25	in.
5216	Cathode illumination	All		---	---	---	---	---	---
5246	Focusing voltage at modulation condition	P1		---	---	Eb1	300	515	Vdc
		P2, P7, P14		---	---	Eb1	400	690	Vdc
5248	Deflection factor	P1	1D2; Eb2 = Eb3 = 1,500 Vdc	---	---	DF	32.5	39.5	Vdc/in.
		P2, P7, P14	1D2, Eb2 = Eb3 = 2,000 Vdc	---	---	DF	43	53	Vdc/in.
5248	Deflection factor	P1	3D4, Eb2 = Eb3 = 1,500 Vdc	---	---	DF	24.5	30.5	Vdc/in.
		P2, P7, P14	3D4; Eb2 = Eb3 = 2,000 Vdc	---	---	DF	32.5	40.5	Vdc/in.
5248	Deflection-factor uniformity	All		---	---	---	---	2.0	%
1216	Base material insulation quality	All		---	---	---	---	---	---
	Quality conformance inspection, part 1								
5106	Bulb, screen, and face-plate quality	All		---	---	---	---	---	---
5201	Voltage breakdown	All		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)	All		---	---	---	---	---	---
5206	Gas "cross"	P1	Light = 15 ftL	---	---	---	---	---	---
		P2, P7, P14	Ib3 = 50 μAdc (see note 1)	---	---	---	---	---	---
5221	Light output	P1	See note 1	---	---	Light	15	---	ftL
5223	Modulation	P1	Light = 15 ftL (see note 1)	---	---	ΔEc	---	45	Vdc
5223	Modulation	P2, P7, P14	Ib3 = 50 μAdc (see note 1)	---	---	ΔEc	---	55	Vdc

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued								
5231	Spot position (electrostatic deflection)	All		---	---	---	---	16	mm
5231	Spot displacement (leakage)	All		---	---	displ	---	10	mm
5241	Grid-cutoff voltage	P1		---	---	Eco	-34	-56	Vdc
		P2, P7, P14		---	---	Eco	-45	-75	Vdc
5251	Grid No. 1 leakage current			---	---	---	---	---	---
5251	Grid No. 2 leakage current			---	---	---	---	---	---
---	Total scan	P1	Focused raster light = 15 ftL (see note 2)	---	---	---	4.25	---	in.
		P2, P7, P14	Focused raster Ib3 = 50 μAdc (see note 2)	---	---	---	4.25	---	in.
(E) 5103	Pattern distortion	All	See note 4	---	---	---	---	---	---
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	All		---	---	---	---	---	---
1301	Heater or filament current	All		---	---	If	540	660	mA
5201	Electrode currents (anode No. 1)	All	Ec1 = 0	---	---	Ib1	-15	10	μAdc
5201	Electrode currents (cathode)	P1	Light = 15 ftL	---	---	Ik	---	1,000	μAdc
5201	Electrode currents (cathode)	P2, P7, P14	Ib3 = 50 μAdc	---	---	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	All	+1D2, pin No. 5	---	---	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	+1D2	---	---	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All		---	---	---	---	---	---
5101	Angle between traces	All		---	---	---	89	91	Degrees
5216	Stray light emission (conventional types)	All	Eb2 = 2,850 Vdc; Eb3 = 6,600 Vdc	---	---	---	---	---	---
5221	Screens	P2, P7, P14	See note 1	---	---	---	---	---	---
(E) 1111	Base pin solder depth	All		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
5226	Line width A (electrostatic deflection)	P1	Light = 15 ftL	---	---	width	---	0.75	mm
		P2, P7, P14	Ib3 = 50 µAdc	---	---	width	---	0.8	mm
5226	Line width B (electrostatic deflection)	P1	Light = 15 ftL (see note 3)	---	---	width	---	0.8	mm
		P2, P7, P14	Ib3 = 50 µAdc	---	---	width	---	0.9	mm
5246	Focusing voltage at cutoff	P1		---	---	Eb1	345	515	Vdc
		P2, P7, P14		---	---	Eb1	460	690	Vdc
5248	Deflection factor	P1	1D2	---	---	DF	40	50	Vdc/in.
		P2, P7, P14	1D2	---	---	DF	54	66	Vdc/in.
5248	Deflection factor	P1	3D4	---	---	DF	30.5	37.5	Vdc/in.
		P2, P7, P14	3D4	---	---	DF	40.5	50	Vdc/in.
5251	Heater-cathode leakage current	All		---	---	---	---	---	---
1105	Permanence of marking	All		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life-test provisions	All	Group C, Eb2 = 2,850 Vdc, Eb3 = 6,600 Vdc, t = 500 hours	---	---	---	---	---	---
		P1	Light = 15 ftL	---	---	---	---	---	---
		P2, P7, P14	Ib3 = 30 µAdc	---	---	---	---	---	---
---	Life-test end points	P1	Light = 11 ftL	---	---	---	---	---	---
			Line width A	---	---	width	---	0.75	mm
			Line width B	---	---	width	---	0.85	mm
			Modulation	---	---	ΔEc	---	45	Vdc
			Heater-cathode leakage current	---	---	---	---	---	---
			Grid No. 1 leakage current	---	---	---	---	---	---
			Grid No. 2 leakage current	---	---	---	---	---	---
			Stray light emission	---	---	---	---	---	---
		P2, P7, P14	Ib3 = 37.5 µAdc	---	---	---	---	---	---
			Line width A	---	---	width	---	0.8	mm
			Line width B	---	---	width	---	0.9	mm
			Modulation	---	---	ΔEc	---	55	Vdc
			Heater-cathode leakage current	---	---	---	---	---	---
			Grid No. 1 leakage current	---	---	---	---	---	---
			Grid No. 2 leakage current	---	---	---	---	---	---
			Stray light emission	---	---	---	---	---	---

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. The scan from the tube face center to the extinction points of focused raster shall be ±2-1/8 inches (53.98 mm).
3. Measure line width B at a distance from the center of the screen equal to 1/3 of the maximum bulb diameter. The applied astigmatism voltage shall be equal to zero volt.
4. With a raster pattern the size of which is adjusted so that the widest points of the pattern just touch the sides of a square 3.075 inches (78.11 mm) on a side, no point on these pattern sides will lie within a square 2.925 inches (74.30 mm) on a side, and having the same center as the first square.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, CE, SM, MU
Navy -
Air Force - 11, 80
DSA - ES

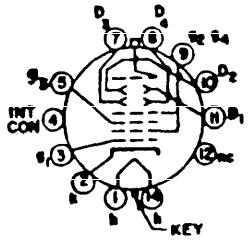
User activities:
Army - MC
Navy - AS, OS, MC, CG, SH
Air Force - None

Preparing activity: Army - EL

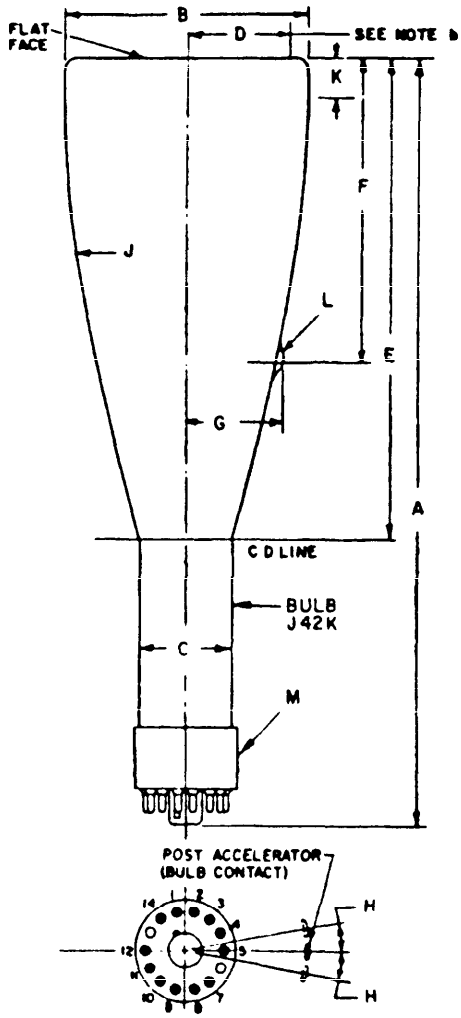
Agent: DSA - ES

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BASING DIAGRAM



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
L	Bulb contact: J1-22	
M	Base: B12-37	
Quality conformance inspection, Part 2		
A	16.563 (420.70)	16.938 (430.23)
B	5.156 (130.96)	5.344 (135.75)
C	1.938 (49.23)	2.063 (52.40)
D	2.250R (57.15)	
F	6.000 (152.40)	6.500 (165.10)
G	2.063 (52.40)	2.563 (65.10)
H		10°
Reference Dimensions (see note c)		
E	[illegible]	[illegible]
J	[illegible]	[illegible]
K	[illegible]	[illegible]

NOTES:
a. Dimensions are in inches.
b. Useful screen radius.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5ADP1, 5ADP2, 5ADP7, and 5ADP14.

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This she is provided for obtaining information on the use of this specification which will insure that suitable produ can be procured with a minimum amount of delay and at the least cost. Comments and the return of this fo will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/689E, ELECTRON TUBES, CATHODE RAY TYPES 5ADP1, 5ADP2, 5ADP7, AND 5ADP11

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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